Input **micro-C code**

```
1
2  int swap(int *x, int *y) {
3      int temp = *x;
4      *x = *y;
5      *y = temp;
6  }
7
8  int main() {
9      int x = 5;
10     int y = 10;
11     swap(&x, &y);
12     return 0;
13 }
```

compile Load example

```c
int swap(int *x, int *y) {
        int temp = *x;
        *x = *y;
        *y = temp;
}

int main() {
        int x = 5;
        int y = 10;
        swap(&x, &y);
        return 0;
}
```

```asm
swap:
    PUSH %BP
    MOV  %SP, %BP
@swap_body:
    MOV  8(%BP), %0
    MOV  (%0), %0
    SUB  %SP, $4, %SP
    MOV  %0, -4(%BP)
    MOV  12(%BP), %0
    MOV  (%0), %0
    MOV  8(%BP), %1
    MOV  %0, (%1)
    MOV  12(%BP), %0
    MOV  -4(%BP), (%0)
@swap_exit:
    MOV  %BP, %SP
    POP  %BP
    RET
main:
    PUSH %BP
    MOV  %SP, %BP
@main_body:
    SUB  %SP, $4, %SP
    MOV  $5, -4(%BP)
    SUB  %SP, $4, %SP
    MOV  $10, -8(%BP)
    LEA  -4(%BP), %0
    LEA  -8(%BP), %1
    PUSH %1
    PUSH %0
    CALL swap
    ADD  %SP, $8, %SP
    MOV  $0, %13
    JMP  @main_exit
@main_exit:
```

text

```
swap:
0   PUSH  %BP
4   MOV   %SP      %BP
@swap_body:
8   MOV   8(%BP)   %0
12  MOV   (%0)     %0
16  SUB   %SP      $4      %SP
20  MOV   %0       -4(%BP)
24  MOV   12(%BP)  %0
28  MOV   (%0)     %0
32  MOV   8(%BP)   %1
36  MOV   %0       (%1)
40  MOV   12(%BP)  %0
44  MOV   -4(%BP)  (%0)
@swap_exit:
48  MOV   %BP      %SP
52  POP   %BP
56  RET
main:
60  PUSH  %BP
64  MOV   %SP      %BP
@main_body:
```

execute restart

binary decimal hexa

stack

632	-1395027775
628	1696110551
624	-1142886408
620	-276841028
616	-2060495974
612	2085667384
608	1377348405
604	527045476
600	1128511764
596	-986518087
592	-155931384
588	1306777127
584	-1033704152
580	381472675
576	396805490
572	-1708646624
568	-1562396609
564	-1048871024
560	-2045305849
556	-2061477715

stack registers

Base pointer %14	636
Stack pointer %15	636

memory

512	243569170
516	-1365101044
520	2072161734
524	-674608651
528	72035664
532	-1494445302
536	431756045
540	1530491598
544	-59200120
548	804039900
552	1391607230

global variables

none defined

flags

ZF	PF	SF	CF	OF
0	0	0	0	0

registers

%0	924480876
%1	-1703605236
%2	-1785433500
%3	1886893290
%4	296964972
%5	-1029115134
%6	1947347718
%7	1973209672
%8	536110012
%9	873834860
%10	2135326552
%11	-659306780
%12	1108320817
%13	1513386470

PUSH %BP

Saves the state of the base pointer onto the stack and increments the stack pointer.

text

```
    swap:
 0    PUSH   %BP
 4    MOV    %SP      %BP
    @swap_body:
 8    MOV    8(%BP)   %0
12    MOV    (%0)     %0
16    SUB    %SP      $4       %SP
20    MOV    %0       -4(%BP)
24    MOV    12(%BP)  %0
28    MOV    (%0)     %0
32    MOV    8(%BP)   %1
36    MOV    %0       (%1)
40    MOV    12(%BP)  %0
44    MOV    -4(%BP)  (%0)
    @swap_exit:
48    MOV    %BP      %SP
52    POP    %BP
56    RET
    main:
60    PUSH   %BP
64    MOV    %SP      %BP
    @main_body:
```

execute restart

binary decimal hexa

stack

632	10101100110110011001000011000001
628	01100101000110001001011111010111
624	10111011111000001110111111111000
620	11101111011111111011110110111100
616	10000101001011110101001110011010
612	01111100010100001100001000111000
608	01010010000110001010101100110101
604	00011111011010100001001101100100
600	01000011010000111011100100010100
596	11000101001100101110110110111001
592	11110110101101001010110100001000
588	01001101111000111101011000100111
584	11000010011000101110110100101000
580	00010110101111001100111110100011
576	00010111101001101100010101110010
572	10011010001010000001111100100000
568	10100010110111111011100000111111
564	11000001011110110111111110010000
560	10000110000101110001110000000111
556	10000101001000000101100010101101

stack registers

Base pointer %14	636
Stack pointer %15	636

memory

512	00001110100001001001001000010010
516	10101110101000100011011000001100
520	01111011100000101010110111000110
524	11010111110010100100100111110101
528	00000100010010110010110101010000
532	10100110111011001001001100001010
536	00011001101111000001001100001101
540	01011011001110010111001011001110
544	11111100011110001010110110001000
548	00101111111011001010110011011100
552	01010010111100100011110110111110

global variables

none defined

flags

ZF	PF	SF	CF	OF
0	0	0	0	0

registers

%0	00110111000110100111010101101100
%1	10011010011101010000110000001100
%2	10010101100101000111001001100100
%3	01110000011101111011010011101010
%4	00010001101100110101001101101100
%5	11000010101010001111001100000010
%6	01110100000100100010101100000110
%7	01110101100111001100101001001000
%8	00011111111101000110001110111100
%9	00110100000101011010010101101100
%10	01111111010001100111111101011000
%11	11011000101100111100011011100100
%12	01000010000011111010001000110001
%13	01011010001101000111000111100110

LEA -4(%BP),%0

Loads the effetive address of the operand into a next free working register.

```
44    MOV    -4(%BP)    (%0)
      @swap_exit:
48    MOV    %BP        %SP
52    POP    %BP
56    RET
      main:
60    PUSH   %BP
64    MOV    %SP        %BP
      @main_body:
68    SUB    %SP        $4        %SP
72    MOV    $5         -4(%BP)
76    SUB    %SP        $4        %SP
80    MOV    $10        -8(%BP)
84    LEA    -4(%BP)    %0
88    LEA    -8(%BP)    %1
92    PUSH   %1
96    PUSH   %0
100   CALL   swap
104   ADD    %SP        $8        %SP
108   MOV    $0         %13
112   JMP    @main_exit
      @main_exit:
116   MOV    %BP        %SP
```

execute restart

binary decimal hexa

stack	
632	00000000000000000000001001111100
628	00000000000000000000000000000101
624	00000000000000000000000000001010
620	11101111011111111011110110111100
616	10000101001011110101001110011010
612	01111100010100001100001000111000
608	01010010000110001010101100110101
604	00011111011010100001001101100100
600	01000011010000111011100100010100
596	11000101001100101110110110111001
592	11110110101101001010110100001000
588	01001101111000111101011000100111
584	11000010011000101110110100101000
580	00010110101111001100111110100011
576	00010111101001101100010101110010
572	10011010001010000001111100100000
568	10100010110111111011100000111111
564	11000001011110110111111110010000
560	10000110000101110001110000000111
556	10000101001000000101100010101101

stack registers	
Base pointer %14	632
Stack pointer %15	624

memory	
512	00001110100001001001001000010010
516	10101110101000100011011000001100
520	01111011100000101010110111000110
524	11010111110010100100100111110101
528	00000100010010110010110101010000
532	10100110111011001001001100001010
536	00011001101111000001001100001101
540	01011011001110010111001011001110
544	11111100011110001010110110001000
548	00101111111011001010110011011100
552	01010010111100100011110110111110

global variables
none defined

flags				
ZF	PF	SF	CF	OF
0	1	0	0	0

registers	
%0	00110111000110100111010101101100
%1	10011010011101010000110000001100
%2	10010101100101000111001001100100
%3	01110000011101111011010011101010
%4	00010001101100110101001101101100
%5	11000010101010001111001100000010
%6	01110100000100100010101100000110
%7	01110101100111001100101001001000
%8	00011111111101000110001110111100
%9	00110100000101011010100101101100
%10	01111111010001100111111101011000
%11	11011000101100111100011011100100
%12	01000010000011111010001000110001
%13	01011010001101000111000111100110

MOV %SP,%BP

Move the value of the stack pointer to the base pointer. Let the base pointer point to the current top of the stack (stack pointer always points to the top of the stack).

```
32    MOV   8(%BP)    %1
36    MOV   %0        (%1)
40    MOV   12(%BP)   %0
44    MOV   -4(%BP)   (%0)
@swap_exit:
48    MOV   %BP       %SP
52    POP   %BP
56    RET
main:
60    PUSH  %BP
64    MOV   %SP       %BP
@main_body:
68    SUB   %SP       $4      %SP
72    MOV   $5        -4(%BP)
76    SUB   %SP       $4      %SP
80    MOV   $10       -8(%BP)
84    LEA   -4(%BP)   %0
88    LEA   -8(%BP)   %1
92    PUSH  %1
96    PUSH  %0
100   CALL  swap
104   ADD   %SP       $8      %SP
108   MOV   $0        %13
```

execute restart

binary decimal hexa

stack	
632	636
628	248822269
624	1222860747
620	677232606
616	1253430998
612	-1138508998
608	-1631923550
604	1021289170
600	1214183386
596	1296922235
592	-676968156
588	-1731758521
584	-761273210
580	556011982
576	1485518315
572	876059657
568	820832166
564	-806036327
560	-1792368805
556	913612641

stack registers	
Base pointer %14	636
Stack pointer %15	632

memory	
512	2139699793
516	-1396495798
520	1235268032
524	-324139874
528	-1561116161
532	-183676803
536	45648481
540	1529830224
544	974500796
548	-582094846
552	-2081110336

global variables
none defined

flags				
ZF	PF	SF	CF	OF
0	0	0	0	0

registers	
%0	1879576732
%1	715362781
%2	-1453800745
%3	-360168506
%4	1689229366
%5	-1292936558
%6	782849858
%7	1716804388
%8	1554386997
%9	-1906045777
%10	553544738
%11	-2080426592
%12	-972549303
%13	-1402330006